AGH University of Science and Technology

Fourier Series (Part 1)
Scalar product, norm, orthogonality...
General formulae
for Electronics and Telecommunications

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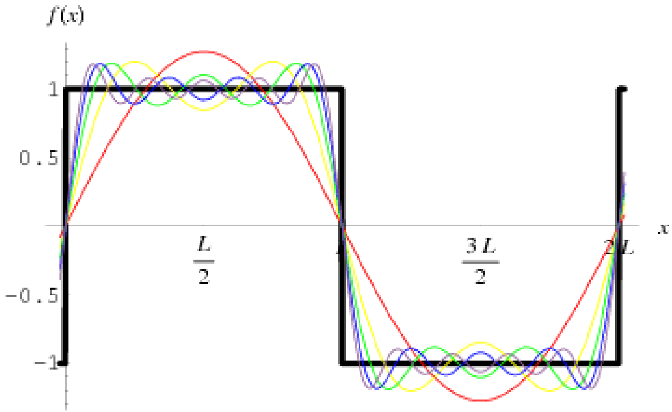
Nice picture, hard theory :-)



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The Aim

The aim of this introduction to Fourier Analysis is to make it simple and avoid advanced notions from functional analysis, while keeping quoted definitions and results rigorous.
At the same time informal comments are given comparing the infinite dimensional setting with finite dimensional vector space (eg. $\mathbb{R}^2$).
However, it is impossible to keep this analogy, when different notions of convergence come to play - pointwise, square mean and uniform.
Especially square mean convergence should be of interest, in the view of further studies.

Outline (Part 1)

- Scalar product in the space of integrable functions,
- Quadratic norm (*rms- root mean square*) and Schwarz inequality,
- Orthogonal and orthonormal sequence and series,
- The Euler-Fourier Theorem.

Assumption for the whole lecture

All functions under consideration are square integrable on $\langle a, b\rangle$.
The vector space consisting of those functions is denoted by $L^2(\langle a, b\rangle)$.

Scalar Product

In some vector spaces we can define a scalar product (with which you are familiar in $\mathbb{R}^2$). By applying scalar product one can chack if two vectors are orthogonal and construct an orthogonal base in $\mathbb{R}^2$.

Scalar Product

Given two vectors f and h their scalar product is a non-negative number $\langle f, h \rangle$ which satisfies properties

- **positively defined**: $\langle f, f \rangle > 0$, for any $f > 0$
- **symmetry**: $\langle f, h \rangle = \langle h, f \rangle$
- **scalar multiplicative**: $\langle \alpha f, h \rangle = \alpha \langle f, h \rangle$
- **distributive law**: $\langle f, h_1 + h_2 \rangle = \langle f, h_1 \rangle + \langle f, h_2 \rangle$

The latter two are called **bilinearity**.

Scalar Product of functions

Scalar Product in L^2

Given two square integrable functions f and h defined on $\langle a, b\rangle$ the number

$$\langle f, h\rangle := \int_a^b f(x)h(x)\mathrm{dx}$$

is their scalar product.

Check if the properties from the previous slide are satisfied.

Orthogonal Functions

Orthogonal Functions

We say that the functions f and h are orthogonal on the interval $\langle a, b\rangle$ iff $\langle f, h\rangle = 0$.

Example.
$f(x) = x, \quad h(x) = x^2$ are orthogonal at $\langle -1, 1\rangle$,
but are NOT orthogonal at $\langle 0, 1\rangle$.

The Norm

Norm

The norm of the vector f is a nonnegative number $\|f\|$ which satisfies conditions:

1.) $\|f\| = 0 \;\Leftrightarrow\; f = 0$;

2.) $\|\alpha f\| = |\alpha|\|f\|$, where $\alpha \in \mathbb{R}$;

3.) $\|f + g\| \leq \|f\| + \|g\|$, for any two vectors f, g.

The Norm of a Square Integrable Function

In L^2 we denote the norm of the function f by $\|f\|_2$ instead of $\|f\|$.

Quadratic Norm

Quadratic norm of the function f on $\langle a, b\rangle$:

$$\|f\|_2 := \sqrt{\int_a^b f^2(x)\mathrm{dx}} \qquad \left(= \sqrt{\langle f, f\rangle}\right)$$

Schwarz inequality $\boxed{\langle f, g\rangle \leq \|f\|_2 \cdot \|g\|_2}$

Remark.

If f is a continuous function on the interval $\langle a, b\rangle$ then $\|f\|_2 = 0 \;\Leftrightarrow\; f \equiv 0$. This norm is not going to distinguish between functions, which differ at the sets of zero measure.

Metric Induced by a Norm

Once we have a vector space equipped with a scalar product we can measure the length of vectors by a norm (induced by our scalar product), and the distance between the vectors by the metric (induced by the norm). This is true for any vector space (eg. $\mathbb{R}^2$), but now we are interested in the space of square integrable functions.

The Metric

The Metric for Square Integrable Functions induced by the L^2 norm

The distance between the vector f and g is measured by a nonnegative number $d_2(f,g)$ defined as $d_2(f,g) := \|f-g\|_2^2$. The function d_2 is called the metric.

The Mean Square Convergence

In an abstract vector space we can not talk about convergence untill we have defined some measure of vectors being close. In the infinite dimension there are many (non equivalent) ways of measuring how close the vectors are.

Convergence in the Mean Square norm in L^2

We say that the sequance $(f_n)_0^{+\infty}$ of functions from L^2 is mean square convergent to $f \in L^2$ if

$$\|f_n - f\|_2^2 \longrightarrow 0, \text{ if } n \longrightarrow +\infty.$$

The Euler-Fourier Thm states about this type of convergence.

Orthogonal Sequence of Functions

The sequence of functions:

$$(\varphi_n(x)) = \varphi_0(x), \varphi_1(x), \varphi_2(x), ...$$

is called orthogonal on the interval $\langle a, b\rangle$ if:

(1) $\forall m, n \in \mathbb{N}, m \neq n : \langle \varphi_m, \varphi_n \rangle = 0$ (any two are orthogonal)

(2) $\forall : \|\varphi_n\|_2 > 0$ (no zero functions in this sequence)

Orthonormal sequence of functions

Orthonormal sequence of functions

Sequence of the functions $(\varphi_n(x))$ is called orthonormal if:

$$\langle \varphi_n, \varphi_m \rangle = \delta_{nm} = \begin{cases} 0 & \text{if } n \neq m \\ 1 & \text{if } n = m \end{cases}$$

(δ_{nm} is called Kronecker delta).

In other words, the sequence of functions is orthogonal and the norm of each of those functions is 1, then it is orthonormal.

Orthogonal Series

Orthogonal Series

Let $(\varphi_n(x))$ be an orthogonal sequence on $\langle a, b\rangle$ and (γ_n) be any number sequence then

$$\sum_{n=0}^{\infty} \gamma_n \varphi_n(x)$$

is an orthogonal series*.

* It should be formally written as $\sum_{n=0}^{\infty} \gamma_n \varphi_n(\cdot)$, rather then $\sum_{n=0}^{\infty} \gamma_n \varphi_n(x)$ (but the latter notation is broadly used by engineeres).

Convergence of an orthogonal series

Lemma

If f is square integrable on $\langle a, b\rangle$ and the following orthogonal series is converging in the mean square norm on the interval $\langle a, b\rangle$ to f ie.

$$\int_a^b \Big(\sum_{n=0}^{N} \gamma_n \varphi_n(x) - f(x)\Big)^2 dx \underset{N\to+\infty}{\longrightarrow} 0, \text{ then } \gamma_n = \frac{\langle f, \varphi_n\rangle}{\|\varphi_n\|_2^2}.$$

In a different notation same can be written as

$$d_2\Big(\sum_{n=0}^{N} \gamma_n \varphi_n, f\Big) \underset{N\to+\infty}{\longrightarrow} 0 \text{ then } \gamma_n = \frac{\langle f, \varphi_n\rangle}{\|\varphi_n\|_2^2}.$$

Possible changes in the Lemma

1.) Instead of talking about convergence in d_2 metric we can equivalently take the following metric

$$\rho_2(\sum_{n=0}^{N} \gamma_n \varphi_n, f) = \sqrt{\frac{1}{b-a} \int_a^b \left[\sum_{n=0}^{N} \gamma_n \varphi_n(x) - f(x) \right]^2 dx}.$$

2.) We may assume uniform convergence, but not pointwise, but by that we exclude non-smooth functions from the Euler-Fourier Thm (eg. having discontinuities).

Proof

From the assumption we know that

$$f(x) = \sum_{n=1}^{\infty} \gamma_n \varphi_n(x) \quad \Big| \cdot \varphi_m(x)$$

$$f(x)\varphi_m(x) = \sum_{n=1}^{\infty} \gamma_n \varphi_n(x)\varphi_m(x)$$

and once we integrate LHS and RHS

$$\int_a^b f(x)\varphi_m(x)\mathrm{dx} = \int_a^b \sum_{n=1}^{\infty} \gamma_n \varphi_n(x)\varphi_m(x)\mathrm{dx}$$

The LHS is equal to $\langle f, \varphi_m \rangle$, therefore it is enough to show that RHS is equal to $\gamma_m \|\varphi_m\|_2^2$.

Proof continued...

Due to the mean square convergence of the series we can reverse order of summing and integration in the RHS

$$\int_a^b \sum_{n=1}^{\infty} \gamma_n \varphi_n(x)\varphi_m(x)\mathrm{dx} = \sum_{n=1}^{\infty} \int_a^b \gamma_n \varphi_n(x)\varphi_m(x)\mathrm{dx}$$

Proof continued...

Because the series is orthogonal, it means that $\int_a^b \varphi_n(x)\varphi_m(x)\mathrm{dx} = 0$ if $n \neq m$ and the only non-zero value in the RHS sum is when $n = m$, and so

$$\sum_{n=1}^{\infty} \gamma_n \int_a^b \varphi_n(x)\varphi_m(x)\mathrm{dx} = \gamma_m \int_a^b \varphi_m(x)\varphi_m(x)\mathrm{dx} = \gamma_m \|\varphi_m\|_2^2$$

where the last equation follows from the definition of the quadratic norm.

The Euler-Fourier Thm (uniqueness of the representation of certain f by the orthogonal series)

The Euler-Fourier Thm/ Expansion Thm

Given an orthonormal sequence (φ_n) on $\langle a, b\rangle$ for any square integrable function f on $\langle a, b\rangle$, s.t. $\boxed{\star}$ $\|f\|_2^2 = \sum_{n=0}^{+\infty} |\langle f, \varphi_n\rangle|^2$ we can build the orthonormal series with coefficients from the **Euler-Fourier formula**

$$\boxed{\star\star} \quad \sum_{n=0}^{+\infty} c_n\varphi_n, \text{ where } c_n := \langle f, \varphi_n\rangle .$$

Under condition $\boxed{\star}$ the series $\boxed{\star\star}$ is mean square convergent to f.

Important Remark. If we skip condition $\boxed{\star}$ the series from the Euler-Fourier Thm does *not* necessarily need to be convergent, and even if convergent does not need to converge to f. This depends solely on the completeness of the system.

Definition of the Fourier Series

Fourier Series for an *orthogonal* system (φ_n)

The orthogonal series is called the **Fourier series** of f according to (φ_n) on the interval $\langle a, b\rangle$ ie. for any $x \in \langle a, b\rangle$

$$f(x) \sim \sum_{n=1}^{\infty} c_n \varphi_n(x), \text{ where } c_n = \frac{\langle f, \varphi_n\rangle}{\|\varphi_n\|_2^2}.$$

So, for any f we can build a Fourier series (by using Euler-Fourier coefficients) according to any orthogonal sequence (φ_n).
But how is it related to the function f??

What does the sign $\sim$ mean?

Important questions and remarks

1. Is the series $\sum_{n=1}^{\infty} c_n \varphi_n(\cdot)$ mean square convergent?
2. If it is mean square convergent, does it converge to the function f used in formula $c_n = \frac{\langle f, \varphi_n \rangle}{\|\varphi_n\|_2^2}$?
3. If the answer to the above two questions is positive, then there is uniqueness of the representation of the function by the series (due to the Lemma preceding the Euler-Fourier Theorem).
4. Under what conditions the series $\sum_{n=1}^{\infty} c_n \varphi_n(\cdot)$ is uniformly convergent?

What does the sign $\sim$ denote?

To get the mean square convergence it is enough to assume that f is square integrable. However if we want to get stronger type of convergence, namely uniform convergence, f needs not only to satisfy Dirichlet conditions (which guaratee only pointwise convergence), but also needs to be smooth (ie. of class $\mathcal{C}^1$).
Proof of the mean square convergence is beyond the scope of this short introduction. It requires using the Bessel inequality and the Schwarz inequality for the first derivative of f (which is continuous) and the complex form of the Fourier series.
Any uniformly convergent sequence is also pointwise convergent at any point, but if the sequence is mean square convergent, then it doesn't necessary need to be pointwise convergent at every point, because the mean square norm "doesn't see" the differences at the sets of zero measure.